

Sensitivity of Wireless Chunk Scheduling to Channel Calibration, Prefetch Restrictions, and Recovery Semantics

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Abstract—Chunk scheduling in heterogeneous wireless clusters couples computation, communication, and intermittent participation. We investigate whether link-aware shrinking, stationary-only prefetch, and a completion cutoff retain their benefits when channel assumptions change. An independent ns-3 calibration supplies application-goodput and conditional echo-delay curves to a SimPy master-worker model. All four workload and availability scenarios are evaluated under this mapping, with disjoint comparator-tuning and evaluation seeds, common recovery semantics, paired ablations, and explicit censoring. A separate intervention tests whether retaining transfers after session expiry alters the conclusions. The tuned comparator selects 1 task(s) per allocation. The full policy meets the joint statistical and practical win rule against that comparator in none of the core scenarios. In the combined scenario, its signed gain is -5.25% relative to fixed dispatch and 10.71% relative to compute-only shrinking. Canceling expired transfers improves compute-only scheduling by 22.61%. The minimum completion fraction across policy-scenario cells is 1.00. Prefetch and recovery interventions are reported with signed effects and matched comparisons rather than credited to the complete policy. These findings concern the sensitivity of an instrumented independent-task model; they do not establish physical-device speedups. Stationary lookup calibration, assumed airtime sharing, unloaded echo delay, and synthetic availability limit generalization to mobile deployments.

Index Terms—Wireless computing, heterogeneous scheduling, prefetch, task recovery, simulation validity, sensitivity analysis.

I. INTRODUCTION

A shrinking allocation can reduce the number of tasks stranded on a slow worker near job completion. Prefetch can overlap input transfer and computation, but can also reserve tasks on a worker that subsequently becomes unavailable. Link-quality factors aim to limit exposure to poor communication. These mechanisms interact with channel modeling and recovery: a purported scheduler advantage can instead reflect an unusually pessimistic rate mapping or bandwidth consumed by obsolete assignments.

This study asks three questions. First, do shrinking and link factors outperform a calibrated fixed-size comparator when communication uses a packet-simulator-derived lookup? Second, do restricting prefetch to stationary workers and stopping it at a completion threshold improve performance? Third, does invalidating a task session without canceling its transfers distort comparisons? The contribution is a controlled

mechanism analysis with retained counterexamples, rather than a claim to invent a universally superior scheduler.

The experiments combine four synthetic scenarios, matched mechanism removals, independent comparator tuning, and a recovery intervention. Every scenario is rerun under the expanded mapping; earlier assumed-channel experiments are retained as model-dependence diagnostics and do not supply the primary speedup claim. Completion and horizon-restricted time are reported together. Physical mobility and end-to-end network execution remain outside the inference scope.

II. RELATED WORK AND COMPARISON SCOPE

Guided self-scheduling and factoring reduce allocation granularity as work decreases [1], [2]. Modern dynamic loop-scheduling libraries expose related tradeoffs on heterogeneous processors [3]; chunk reshaping is also studied in heterogeneous data-processing clusters [4]. Consequently, decreasing chunks and capability weighting alone do not establish algorithmic novelty.

HEFT uses task ranking and earliest predicted finish to schedule heterogeneous resources [5]. An independent-task CPU-only HEFT comparison is retained in the auxiliary artefact; its shared-radio execution is an adaptation. It is not evidence of superiority over a complete published distributed system. The main comparisons instead isolate mechanisms in one common instrumented runtime, which permits a narrower causal interpretation.

Mobile crowd-computing and offloading frameworks address deployment-level questions beyond chunk allocation [6], [7], [8], [9], [10]. This study does not reproduce these systems and makes no comparative performance claim against them. Differences in topology, dependencies, task migration, transport, and instrumentation prevent treating a small speculative-copy policy as their faithful substitute.

The MAC analysis of Bianchi [11] and the 802.11 performance anomaly [12] illustrate why a fluid rate table cannot by itself validate contention. ns-3 provides an independent packet-level modeling framework [13], but agreement between two simulators does not establish real-world validity. Optional drift-detection code uses ADWIN [14]; drift detection is excluded from the evaluated policy and is not a contribution claimed here.

III. MODEL AND CHANNEL CALIBRATION

A. Tasks, Workers, and Participation

A coordinator distributes independent indivisible task identifiers to eight workers. Each job has 240 tasks with nominal compute cost averaging ten reference-worker seconds. Compute heterogeneity spans a factor of four while preserving total nominal speed. Input size is proportional to task cost; results contain one tenth of the input volume. Nominal task costs are known, while worker capability is estimated from completed compute intervals. These assumptions favor instrumented workloads and do not cover unknown-size jobs or DAG dependencies.

The scenarios are uniform links, a stationary weak-link group, intermittent participation, and combined weak links, participation changes, and compute slowdowns. Half of the workers belong to the weak group where applicable; half have dynamic participation where applicable. Availability events include temporary loss, permanent departure, and departure followed by return. Signal traces include synthetic correlated fluctuations and dips. The main cells control signal groups directly; geometric random-walk coordinates do not independently validate their radio traces. Stationary designation is supplied to the scheduler. Full parameters are provided in the supplement.

B. Independent Packet-Level Calibration

ns-3 3.44 models an IEEE 802.11a AP and stationary receivers, 20 MHz at 5 GHz, ideal rate adaptation, 16 dBm transmit power, and log-distance loss with exponent three and reference loss 56 dB. Saturated UDP downlink application goodput is observed for five seconds after association; separate trials send 100 echo probes. AP data-frame RSSI excludes neighboring-station ACKs. With no decoded frame, the lookup coordinate is configured receive power, not an observed RSSI. Echo failure remains a failure: it is represented by a declared 400 ms probe timeout, not zero delay.

Training RNG runs 1–3 fit piecewise-linear goodput and conditional echo-delay knots. Runs 4–5 evaluate the same positions. Initial diagnostics also vary receiver count; the expanded low-signal support is single-receiver calibration. There are 190 packet-level runs in total. Holdout at training distances tests stochastic repeatability, not unseen-distance accuracy. No fading, measured interference, TCP, or mobile scheduler is calibrated.

The expanded single-receiver holdout goodput MAE is 0.015 Mbps. Small error at identical stationary training positions reflects limited stochastic variation and does not validate moving or heterogeneous contending stations.

C. Projection into the Scheduler

Every core scenario uses the expanded goodput lookup for data transfer and the training conditional echo-delay lookup for observations. Out-of-support signals are rejected rather than silently extrapolated. Synthetic departure state is combined with nonzero lookup goodput and probe delivery. An active transfer receives its nominal application-goodput rate

divided by the number of active positive-rate transfers. This equal-airtime projection is assumed; it is not a heterogeneous moving-station MAC calibration. Conditional unloaded echo does not reproduce queueing delay under data load. Probes and compute-progress reports are instantaneous and consume no modeled bandwidth.

Thus, the primary experiment recalibrates all core rate and echo mappings while retaining explicit approximations to contention and mobility. It is neither a fully calibrated network simulator nor an end-to-end scheduler executed inside ns-3. In particular, large speedups under the earlier assumed mapping cannot be promoted to measured mobile-cluster gains.

IV. SCHEDULING AND RECOVERY MECHANISMS

Let R denote unassigned task count and U eligible workers. Compute-only EWMA speed is $\hat{s}_i$, initialized equally and updated from local compute duration. The link-aware chunk rule is

$$k_i = \min \left(R, \max \left(1, \left\lceil .3R \frac{\hat{s}_i}{\sum_{j \in U} \hat{s}_j} \ell_i \delta_i \right\rceil \right) \right), \quad (1)$$

with $\ell_i = \text{clip}((\text{RSSI}_i + 85)/25, .25, 1)$ and $\delta_i = \text{clip}(20/\text{RTT}_i, .25, 1)$, using RTT in milliseconds. A matched compute-only policy removes both factors and prefetch while retaining shrinking and recovery. Removing only RSSI or RTT preserves the other factor and the common eligibility test. Small chunks, clipping, and the critical completion path can make a removal change nothing even when the formula differs.

Idle workers have priority over prefetch. The full policy reserves at most one next chunk for a stationary worker while its current chunk computes. Estimated remaining compute must exceed nominal input-transfer duration plus one second. Prefetch stops at 75% unique accepted completion; existing reservations are retained. These settings are hypotheses under test, not established optimal values. The unrestricted-worker and no-cutoff variants remove the corresponding restriction.

Eligibility requires reachability, RSSI at least -80 dBm, and observed RTT at most 150 ms. Three unhealthy probes, four seconds apart, start a 20-second grace period. Expiry advances the session epoch and returns unaccepted task IDs to the pending set. Task-ID deduplication and epoch fencing reject repeated and stale results. The RSSI threshold is an assumed participation rule, not a validated physical-departure detector.

In the primary runtime, expiry also cancels obsolete coordinator transfers. A transfer waiter receives cancellation and cannot later commit its result. Local computation already in progress can continue during disconnection; a stale output is suppressed. A matched intervention retains expired transfers instead. All main schedulers share recovery, cancellation, and result semantics, avoiding a comparison that confounds chunking with reliability.

V. EXPERIMENTAL AND STATISTICAL PROTOCOL

A. Calibration and Evaluation Separation

Fixed-size comparators are tuned on 10 seeds per scenario, distinct from 50 evaluation seeds. The candidate integer sizes

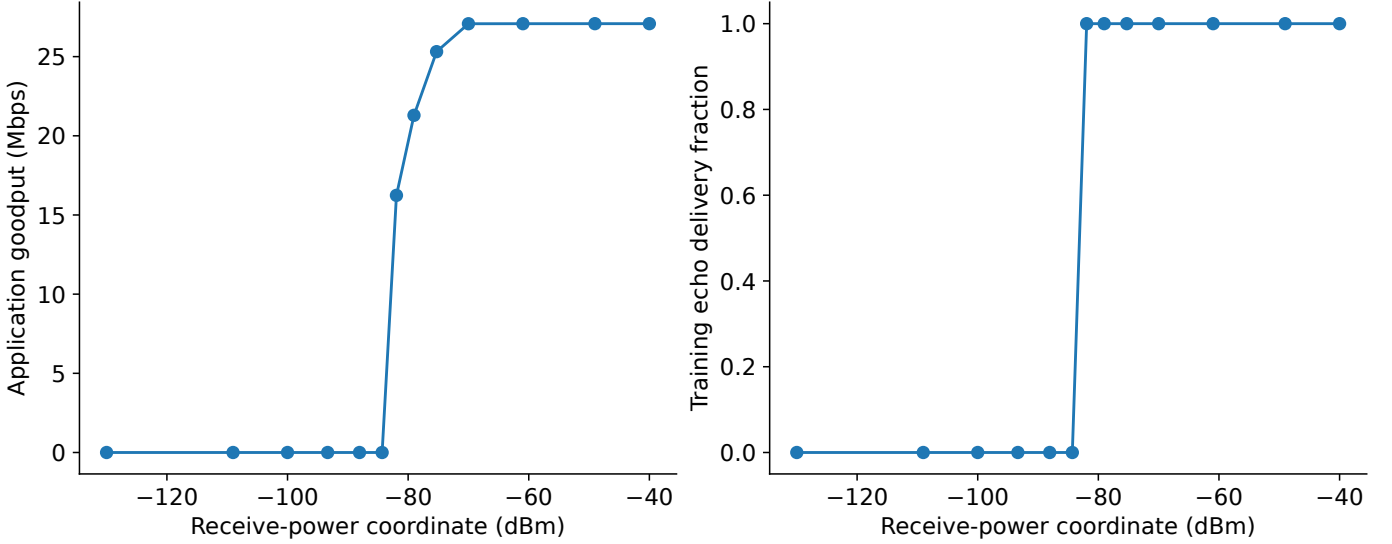


Fig. 1. Expanded ns-3-derived application-goodput lookup and echo delivery. Configured receive power is used where no data frame is decoded. The missing-echo timeout is a protocol convention, not a latency measurement.

are 1, 2, 3, 4, 8, 16, and 32. Minimum worst-scenario mean relative regret selects 1 task(s) per chunk; ties use overall mean then smaller size. One task is the feasible lower boundary: a smaller chunk would require splitting an indivisible task and constitutes a different model. The selected comparator is therefore strong single-task dispatch when that boundary is chosen, not an invalid or fractional baseline.

Evaluation includes the full policy, compute-only shrinking, five mechanism removals/restrictions, the tuned fixed comparator, and two retained-transfer interventions. Each policy uses the same immutable world for a given scenario and seed. These runs total 2000, with 280 additional tuning runs. World generation is independent of scheduling decisions. The administrative horizon is 4,000 seconds. The artefact includes 40 implementation checks and source/configuration hashes. The protocol is exploratory and follows model diagnostics; it is not represented as preregistered.

B. Estimands and Uncertainty

The primary estimand is paired mean $\min(T, H)$, retaining every seed, with completion fractions shown separately. Signed gain is $100(\overline{T}_B^H - \overline{T}_A^H)/\overline{T}_B^H$. A censored run contributes its horizon, not an imputed completion time. This estimand describes performance through a common horizon and cannot establish an unrestricted speedup when jobs remain incomplete.

Paired bootstrap intervals use 5,000 seed resamples. Paired sign-flip tests use 10,000 draws with a plus-one correction and assume sign exchangeability under the null. Holm correction covers 40 comparisons, including four matched compute-policy recovery contrasts. Statistical evidence uses adjusted $p < .05$; an exploratory practical threshold additionally requires at least 1% gain. These are analysis conventions, not a claim of independent prospective specification.

The reported 95% intervals are pointwise bootstrap intervals. They are not simultaneous family-wise intervals or

inversions of the adjusted permutation tests. An interval excluding zero can therefore coexist with a large Holm-adjusted p , including 1.0; resampling and sign-flip assumptions also differ. Interpretation of statistical significance follows adjusted tests, with effect size and completion reported alongside them.

VI. RESULTS

A. Primary Comparators

The tuned comparator selects 1 task(s) per allocation. The full policy meets the joint statistical and practical win rule against that comparator in none of the core scenarios. In the combined scenario, its signed gain is -5.25% relative to fixed dispatch and 10.71% relative to compute-only shrinking. Canceling expired transfers improves compute-only scheduling by 22.61%. The minimum completion fraction across policy-scenario cells is 1.00. Prefetch and recovery interventions are reported with signed effects and matched comparisons rather than credited to the complete policy.

The completion table in the supplement retains every policy and scenario. A favorable contrast against compute-only scheduling does not establish improvement over the stronger tuned comparator.

B. Prefetch and Link-Factor Ablations

Proposed-noRSSI: identical completion times in uniform, churn, full-policy practical wins in weak links, combined.

Proposed-noRTT: identical completion times in uniform, weak links, churn, combined. No full-policy practical win under the adjusted rule.

Proposed-allPrefetch: identical completion times in uniform, weak links. No full-policy practical win under the adjusted rule.

Proposed-noCutoff: signed losses in churn, combined. No full-policy practical win under the adjusted rule.

TABLE I

FULL POLICY VERSUS SEPARATELY TUNED FIXED-SIZE AND MATCHED COMPUTE-ONLY SCHEDULING. POSITIVE VALUES FAVOR THE FULL POLICY.

Scenario	Comparator	Gain (%)	Difference (s)	Pointwise 95% CI (s)	Holm p
uniform	Tuned-fixed	0.01	0.09	[-2.30, 2.48]	1.0000
uniform	Compute-shrink	1.51	18.23	[13.15, 23.29]	0.0040
weak_links	Tuned-fixed	-3.30	-45.70	[-52.44, -38.97]	0.0040
weak_links	Compute-shrink	8.42	131.48	[113.69, 148.98]	0.0040
churn	Tuned-fixed	-1.72	-20.76	[-25.66, -15.80]	0.0040
churn	Compute-shrink	2.86	36.05	[25.53, 48.07]	0.0040
combined	Tuned-fixed	-5.25	-74.30	[-87.00, -62.12]	0.0040
combined	Compute-shrink	10.71	178.77	[158.43, 199.86]	0.0040

TABLE II

MECHANISM CONTRASTS AGAINST THE FULL POLICY. NEGATIVE GAINS FAVOR REMOVING OR RELAXING THE INDICATED MECHANISM.

Scenario	Comparator	Gain (%)	Difference (s)	Pointwise 95% CI (s)	Holm p
uniform	Proposed-noRSSI	0.00	0.00	[0.00, 0.00]	1.0000
uniform	Proposed-noRTT	0.00	0.00	[0.00, 0.00]	1.0000
uniform	Proposed-noPrefetch	1.51	18.23	[13.07, 23.24]	0.0040
uniform	Proposed-allPrefetch	0.00	0.00	[0.00, 0.00]	1.0000
uniform	Proposed-noCutoff	0.15	1.78	[-1.07, 4.72]	1.0000
weak_links	Proposed-noRSSI	8.75	137.23	[121.06, 153.61]	0.0040
weak_links	Proposed-noRTT	0.00	0.00	[0.00, 0.00]	1.0000
weak_links	Proposed-noPrefetch	0.34	4.82	[-2.02, 11.57]	1.0000
weak_links	Proposed-allPrefetch	0.00	0.00	[0.00, 0.00]	1.0000
weak_links	Proposed-noCutoff	0.11	1.63	[-1.55, 4.91]	1.0000
churn	Proposed-noRSSI	0.00	0.00	[0.00, 0.00]	1.0000
churn	Proposed-noRTT	0.00	0.00	[0.00, 0.00]	1.0000
churn	Proposed-noPrefetch	2.86	36.05	[25.40, 48.34]	0.0040
churn	Proposed-allPrefetch	0.10	1.24	[-1.08, 3.55]	1.0000
churn	Proposed-noCutoff	-0.14	-1.69	[-4.82, 1.33]	1.0000
combined	Proposed-noRSSI	10.77	179.80	[156.88, 203.95]	0.0040
combined	Proposed-noRTT	0.00	0.00	[0.00, 0.00]	1.0000
combined	Proposed-noPrefetch	0.85	12.80	[1.08, 25.74]	0.9974
combined	Proposed-allPrefetch	0.21	3.17	[-7.46, 14.33]	1.0000
combined	Proposed-noCutoff	-0.17	-2.49	[-5.69, 0.69]	1.0000

TABLE III

MATCHED COMPUTE-ONLY RECOVERY INTERVENTION: CANDIDATE CANCELS EXPIRED TRANSFERS; COMPARATOR RETAINS THEM.

Scenario	Comparator	Gain (%)	Difference (s)	Pointwise 95% CI (s)	Holm p
uniform	Compute-retainExpired	0.00	0.00	[0.00, 0.00]	1.0000
weak_links	Compute-retainExpired	18.69	359.13	[307.11, 417.79]	0.0040
churn	Compute-retainExpired	1.52	19.53	[14.09, 24.85]	0.0040
combined	Compute-retainExpired	22.61	487.64	[417.38, 560.74]	0.0040

C. Recovery Intervention

The full-policy retained-transfer contrast and all resource counters appear in the supplement. All primary chunking comparisons use cancellation consistently. Requeued IDs, transmitted bytes, and unaccepted computation diagnose simulated overhead, not measured energy or physical failure.

VII. INTERPRETATION AND THREATS TO VALIDITY

The RSSI ablation supports retaining the RSSI factor in the weak-link and combined scenarios, whereas the RTT ablation is identical in every scenario. Under the eligibility threshold and unloaded echo mapping, the RTT chunk factor saturates at one. RSSI benefits may consequently reflect reduced exposure to threshold-triggered invalidation rather

than slow data throughput alone. The result should be read as a mechanism study. Shrinking, link weighting, prefetch, participation restrictions, and session cleanup cannot each be credited with the performance of their combination. Ties and improvements from removals constrain any recommendation. Link-factor inactivity under low conditional RTT is expected from clipping; a negligible gain against single-task dispatch limits claims of extra scheduler value.

Earlier assumed-channel results show that large nominated-baseline gaps can coincide with extensive censoring and repeated task recovery. Extending the same combined worlds increased compute-only completion from 10% to 92%, while leaving four runs censored. That diagnoses strong horizon dependence but does not prove the entire difference is a

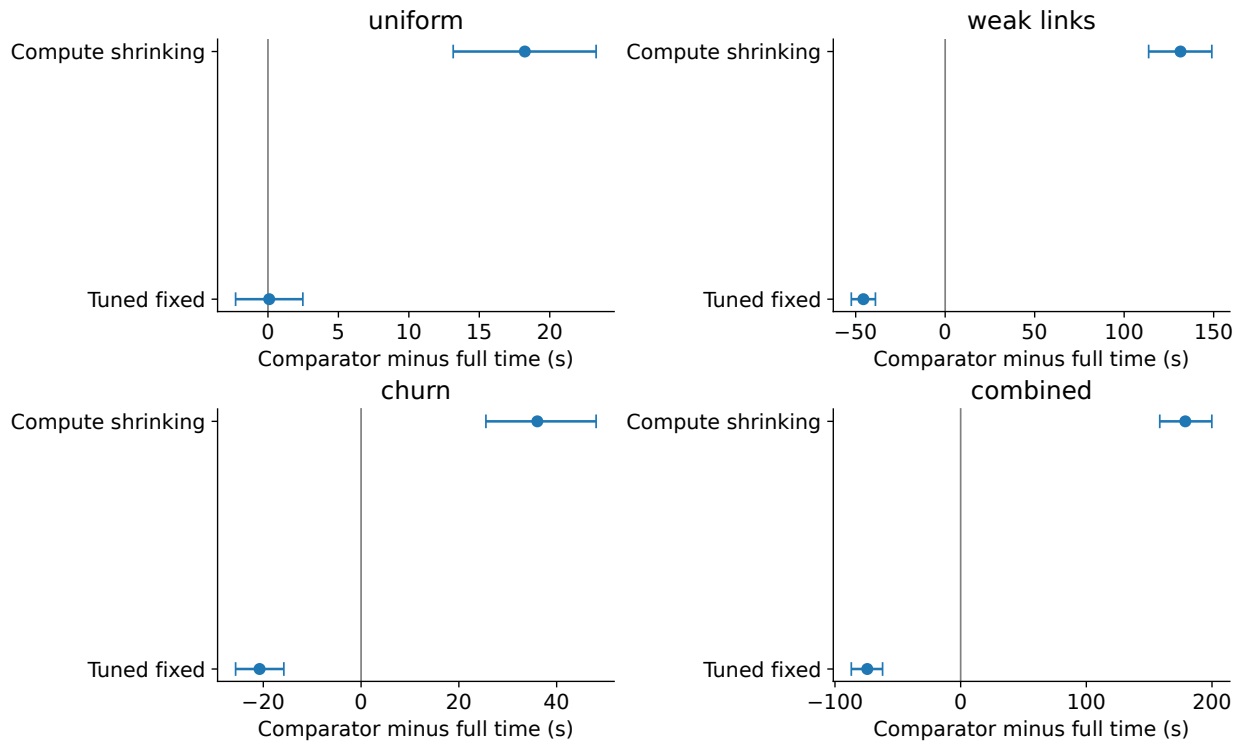


Fig. 2. Primary ns-3-lookup core comparisons against the tuned fixed-size and matched compute-only policies. Positive paired differences favor the full policy. Intervals are pointwise bootstrap intervals.

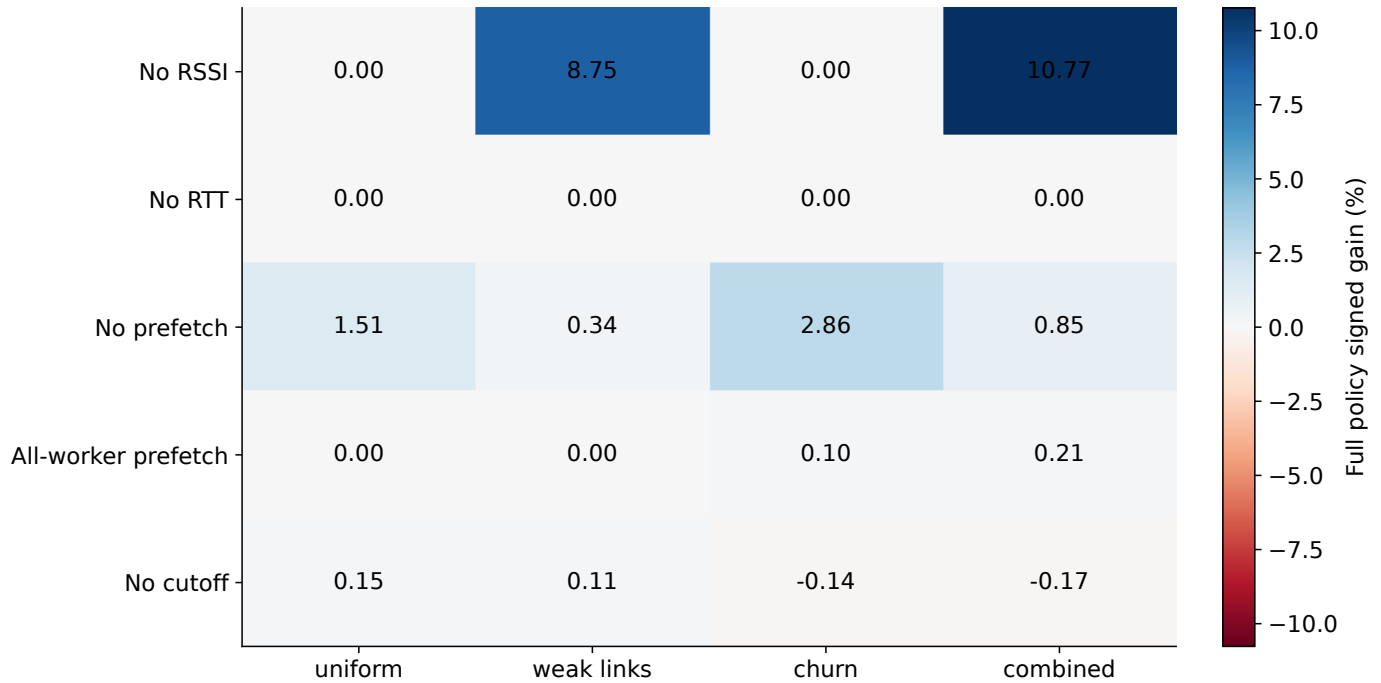


Fig. 3. Signed gains over mechanism variants across all calibrated core scenarios. Color represents effect magnitude, not adjusted statistical significance. Negative values favor removing the mechanism.

censoring artifact. Requests for already assigned data, retained obsolete transfers, and large weak-worker reservations offer concrete simulator-level explanations; the matched cleanup intervention tests one explanation. Primary conclusions use the recalibrated mapping and common runtime, rather than these

earlier percentages.

Calibration remains limited to stationary UDP downlink and conditional unloaded echo. Adapting such curves to changing synthetic signals assumes quasi-stationarity. Equal-airtime allocation, bidirectional behavior, queueing, control

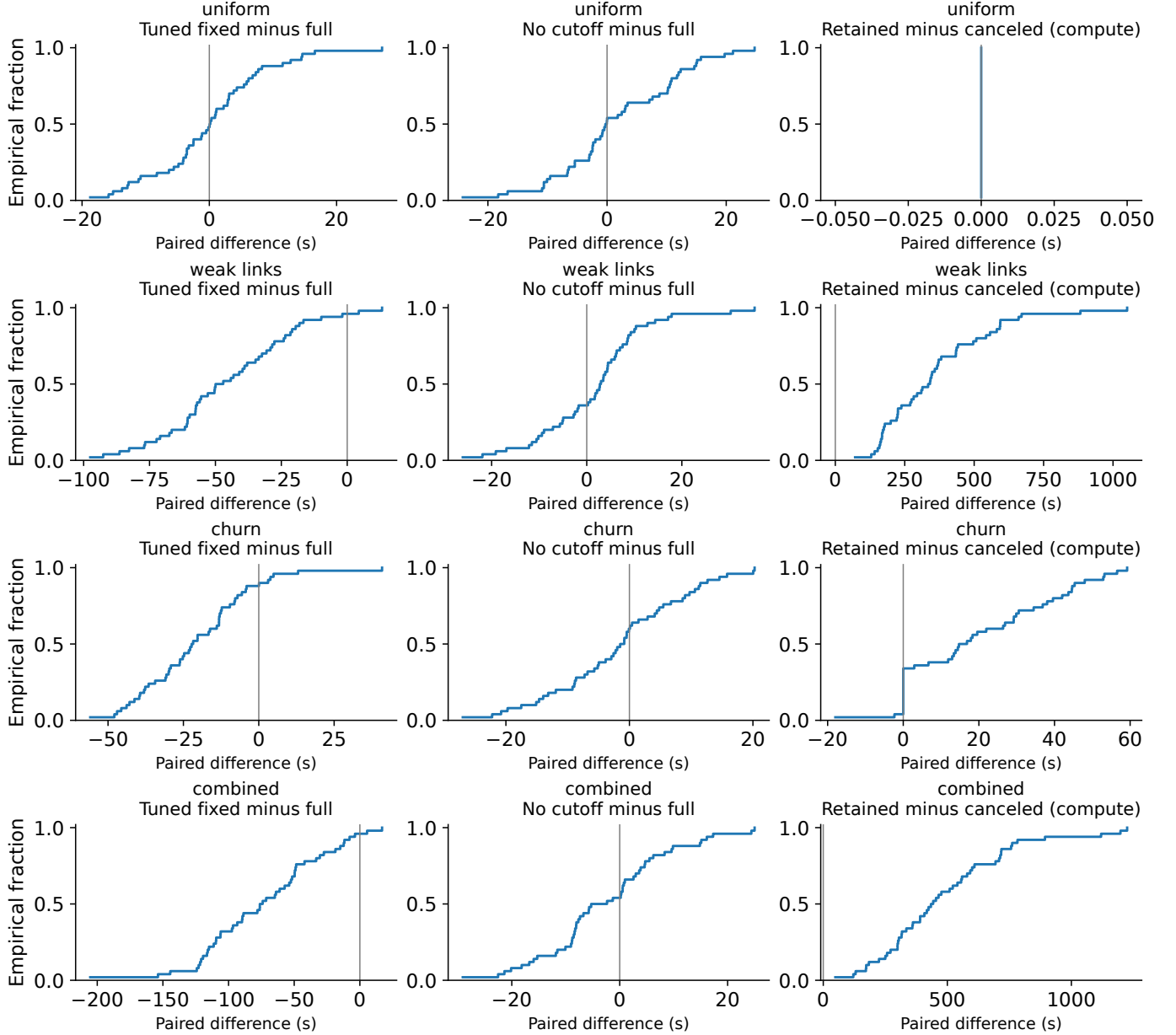


Fig. 4. Dense paired-outcome audit. Every panel contains all 50 evaluation seeds; the vertical line denotes zero difference. Rows retain all scenarios, including ties and losses. Curves describe paired effect distributions and are not separate hypothesis tests.

costs, hardware thermals, and return-session transport are not independently calibrated. The fixed RSSI cutoff may reject a usable radio link and is not proof that a person walked away. The main signal/availability scenarios do not constitute a validated physical mobility model.

The study lacks a measured testbed and a faithful close mobile-framework competitor. It cannot establish deployment effectiveness, superiority over published systems, or independent algorithmic novelty. A next validation should use synchronized transfer/probe/task traces and repeated stationary, weak-link, departure/return, and combined trials, holding workloads and recovery semantics fixed across scheduling variants. Runtime control and progress overhead must be included. These limitations constrain the application of the results rather than being replaced by source-level tests.

VIII. CONCLUSION

This study evaluates the sensitivity of wireless chunk scheduling to an independent channel lookup, prefetch restrictions, and recovery semantics. All core scenarios are rerun using the expanded ns-3-derived mapping, strong separately tuned comparators, paired ablations, and common reliability behavior. The evidence supports mechanism-specific conclusions and retained failures, not a universal scheduler superiority claim. Physical and coupled mobile validation are required before recommending the policy for deployment.

DATA AND CODE AVAILABILITY

Code, calibration records, all per-run results, analysis scripts, figure sources, and supplementary tables are distributed in the accompanying research artefact at <https://github.com>.

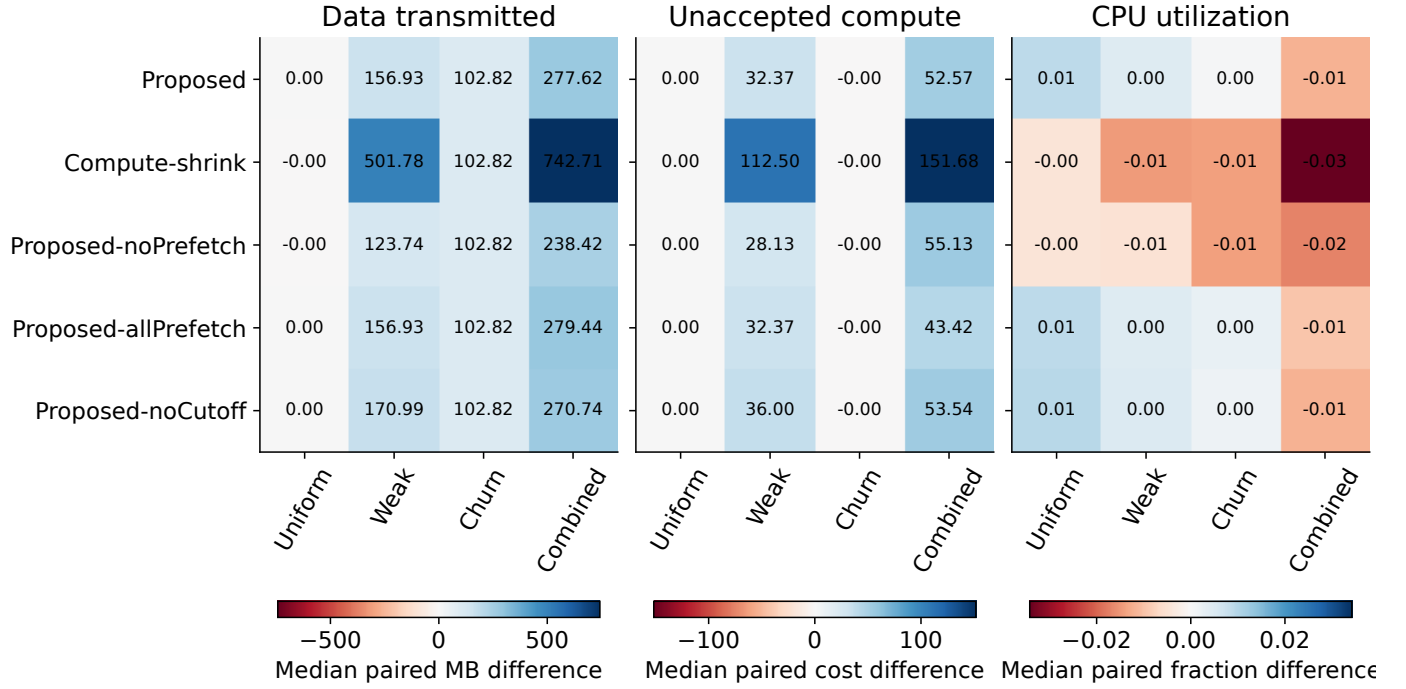


Fig. 5. Descriptive resource audit: median paired differences from tuned fixed-size scheduling. Each metric has its own symmetric color scale and units. Less data or unaccepted compute is favorable; CPU utilization alone does not establish faster completion. These simulator counters are not measured energy or physical airtime.

com/logiqueviper/HPC/tree/research-deliverables. The repository provides reproducibility materials; no archival DOI is asserted. The supplementary material lists full policies, calibration knots, uncertainty estimates, completion rates, and recovery counters.

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